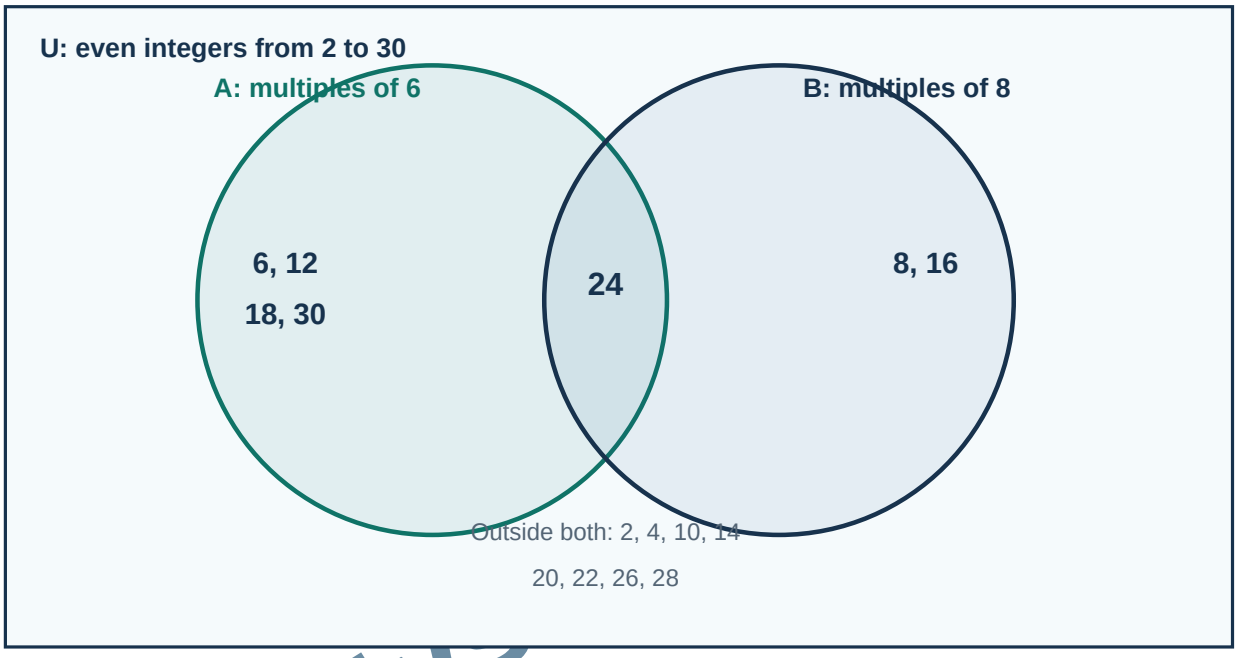
Chapter 3 • Sets and Functions

Solved Exercise 3.2

1. U contains positive even x≤30; A contains multiples of 6 and B multiples of 8.
A = {6,12,18,24,30}; B
= {8,16,24}.
A∩B contains the common element 24.
(i) lists above
(ii) A∩B = {24}
(iii) see the Venn diagram.



2. Within integers 0<x≤150, G = {2ᵐ} and H is the set of squares.
G = {1,2,4,8,16,32,64,128}.
H = {1,4,9,16,25,36,49,64,81,100,121,144}.
G∪H = {1,2,4,8,9,16,25,32,36,49,64,81,100,121,128,144}.
G∩H = {1,4,16,64}.
Lists, union and intersection shown above.

3. P contains primes ≤20 and Q contains positive divisors of 210 not exceeding 20.
P = {2,3,5,7,11,13,17,19}
Q = {1,2,3,5,6,7,10,14,15}.
P∩Q = {2,3,5,7}.
P∪Q = {1,2,3,5,6,7,10,11,13,14,15,17,19}.
Intersection and union shown above.

4. Verify commutativity of union and intersection.
(i) A∪B = B∪A
= {1,2,3,4,5,6,8,10}
A∩B = B∩A
= {4}.
(ii) N∪Z = Z∪N = Z and N∩Z = Z∩N = N.
(iii) If A = {x∈R | x≥0} and B = R, then A∪B = B∪A = R and A∩B = B∩A = A.

5. For the given U, A and B, verify De Morgan's laws.
A∪B = {a,b,c,d,e,f,g,h,j}, so (A∪B)′
= {i}.
A′ = {e,f,i,j}, B′ = {a,b,g,h,i}; hence A′∩B′ = {i}.
A∩B = {c,d}, so (A∩B)′
= {a,b,e,f,g,h,i,j}.
A′∪B′ = {a,b,e,f,g,h,i,j}.

(A∪B)′ = A′∩B′ and (A∩B)′ = A′∪B′.

6. For U = {1,…,20} and A = {1,3,…,19}, verify the three identities.
(ii) A∩U = A
(iii) A∩A′ = ∅.

7. In a class of 55, 34 like cricket and 30 hockey; everyone likes at least one.
n(C∩H) = n(C)+n(H)−n(C∪H)
= 34+30−55
= 9.
9 students like both games.

8. Language survey: n(Urdu)=250, n(English)=150, n(Punjabi)=50, pairwise overlaps 40, 30 and 10; total employees 500.
If all 500 were intended to speak at least one language, inclusion–exclusion would give 500 = 250+150+50−40−30−10+t, so t = 130.
But t cannot exceed n(Punjabi) = 50, nor can it exceed the Urdu–Punjabi overlap 10. Thus t = 130 is impossible and the printed data have no valid three-set Venn diagram.

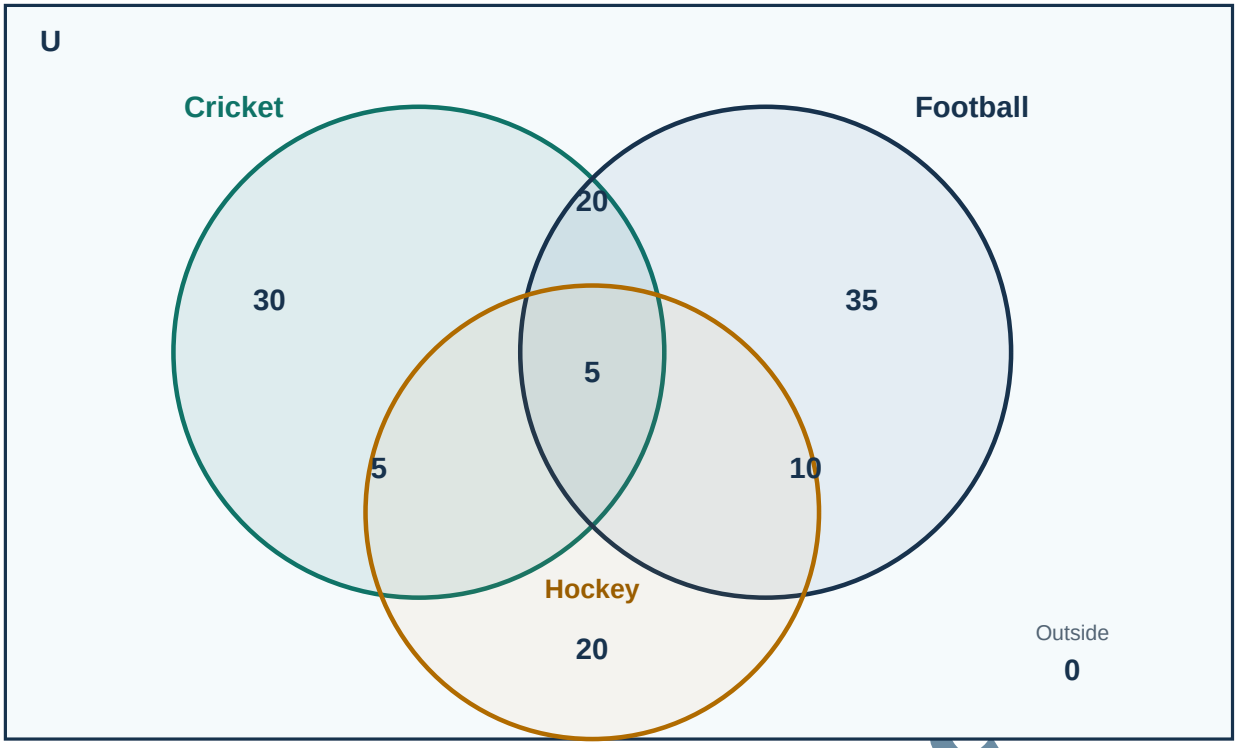
9. Shirt/cap survey: blue 19, green 15, blue∩green 3, cap∩blue 4, cap∩green 2, union 34. Find caps.
Let C be the number wearing caps and t the number in all three categories.
34 = 19+15+C−3−4−2+t, so C
= 9−t.
Under the key's implicit assumption t = 0 (no one is in all three categories), C = 9.
9 caps under the no-triple-overlap assumption; otherwise the data give only C = 9−t.

10. Training session: laptops 17, tablets 11, LT=9, LB=6, TB=4, all three=8, union=35. Find books B.
Formal inclusion–exclusion: 35 = 17+11+B−9−6−4+8, hence B = 18.
However, the stated triple overlap 8 already exceeds LB=6 and TB=4, which is impossible.

11. U = {1,…,150}; A = {50,…,89}, B = {101,…,150}, C = {1,…,49}∪{90,…,100}.
(a) A∩B = ∅, so A′∪B′ = (A∩B)′ = U. Therefore (A′∪B′)∩C = C.
{1,2,…,49,90,91,…,100}
(b) B′ = {1,…,100}; C′ = {50,…,89}∪{101,…,150}. Hence B′∩C′ = {50,…,89} = A.
Thus A∩(B′∩C′) = A, which has 40 elements.
40.

12. All 125 students play cricket, football or hockey. Totals are 60, 70, 40; pairwise overlaps 25, 15, 10.
125 = 60+70+40−25−15−10+t
⇒ t = 5.
Only CF = 25−5 = 20; only FH = 15−5 = 10; only CH = 10−5 = 5.
Only cricket = 60−20−5−5 = 30; only football = 70−20−10−5 = 35; only hockey = 40−5−10−5 = 20.
(a) 5 play all three
(b) the seven regions are shown below.

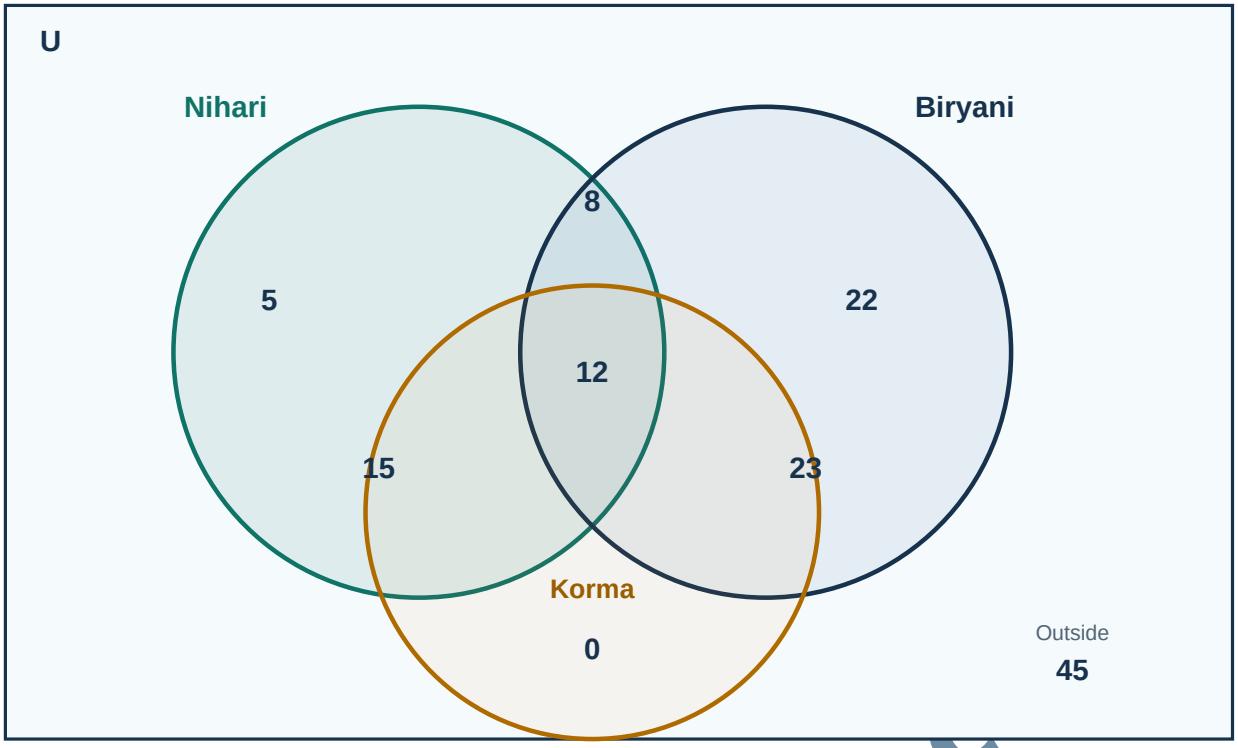
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13. Food survey of 130: nihari 40, biryani 65, korma 50; pairwise overlaps 20, 35, 27; all three 12.

At least one = 40+65+50−20−35−27+12
= 85; none
= 130−85
= 45.
Only NB = 8, only BK = 23, only NK = 15.
Only nihari = 40−8−15−12 = 5; only biryani = 65−8−23−12 = 22; only korma = 50−15−23−12 = 0.
Exactly one food = 5+22+0
= 27.
(a) 85
(b) 45
(c) 27
(d) see the Venn diagram.

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